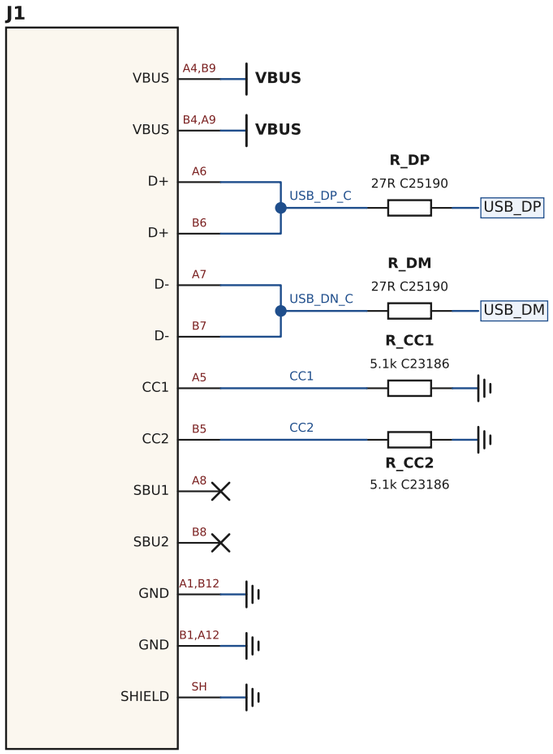
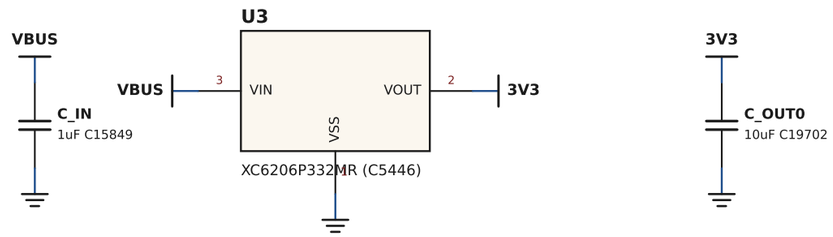
**USB-C (device): 5.1k pull-downs on CC1 and CC2, the D+ and D- pairs joined on the board, so either plug orientation works**

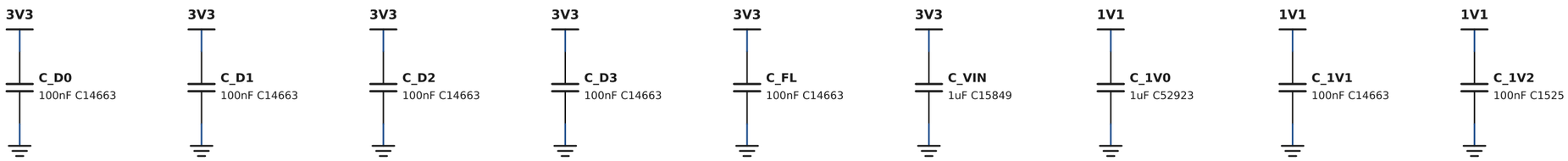
**3.3 V regulation**




TYPE-C-31-M-12 USB-C receptacle (C165948)




**Decoupling, placed at the RP2040 and the flash; the 3V3 and GND inner planes connect them**




C_D0 to C_D3: IOVDD, USB_VDD and ADC_AVDD. C_FL: flash VCC. C_VIN: VREG_IN.

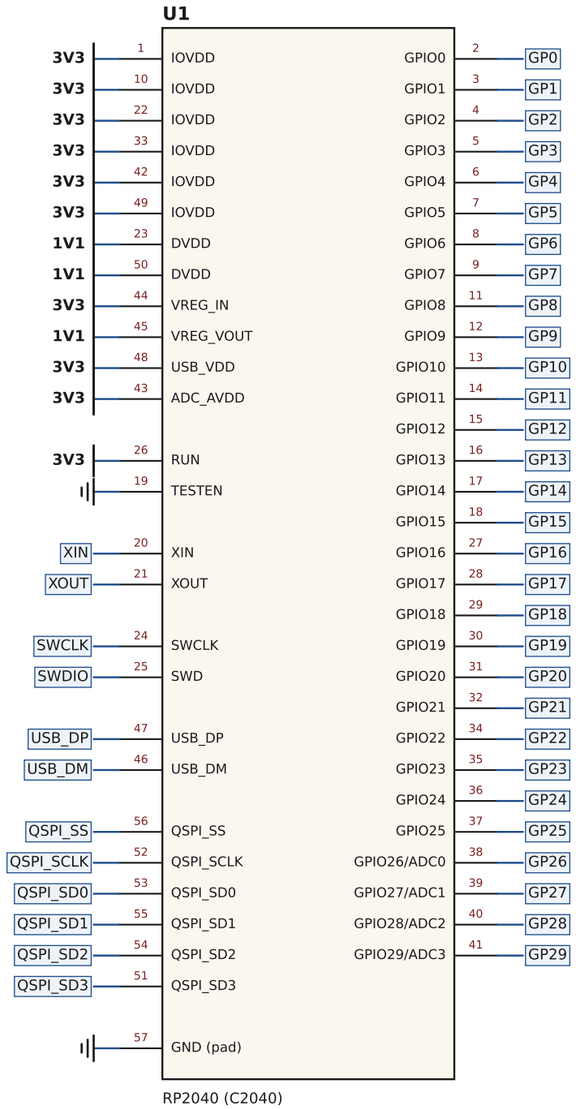
C_1V0 to C_1V2: DVDD and VREG_VOUT (the RP2040's internal 1.1 V core regulator).

**Avionics Panel rev 1, schematic**

Sheet 1 of 4: USB-C and power

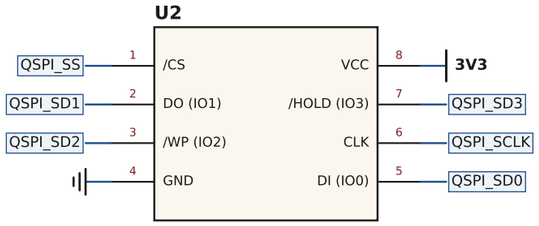
Drawn from the netlist of pcb/board.py (make_schematic.py)

## RP2040, chip-down minimal design (Raspberry Pi, Hardware design with RP2040)

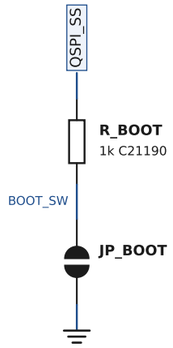



RP2040 (C2040)

## Flash and USB boot

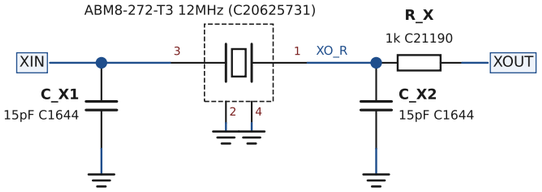



W25Q16JVSNIQ 16 Mbit QSPI flash (C2456211)

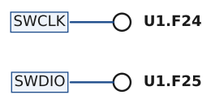



JP_BOOT: bridge pads under a hole in the case floor. Bridged while plugging in USB, chip select low starts the USB bootloader.

## 12 MHz crystal




## SWD debug pads (exposed copper, bottom side)




RUN (pin 26) is tied to 3V3: the probe resets over SWD.

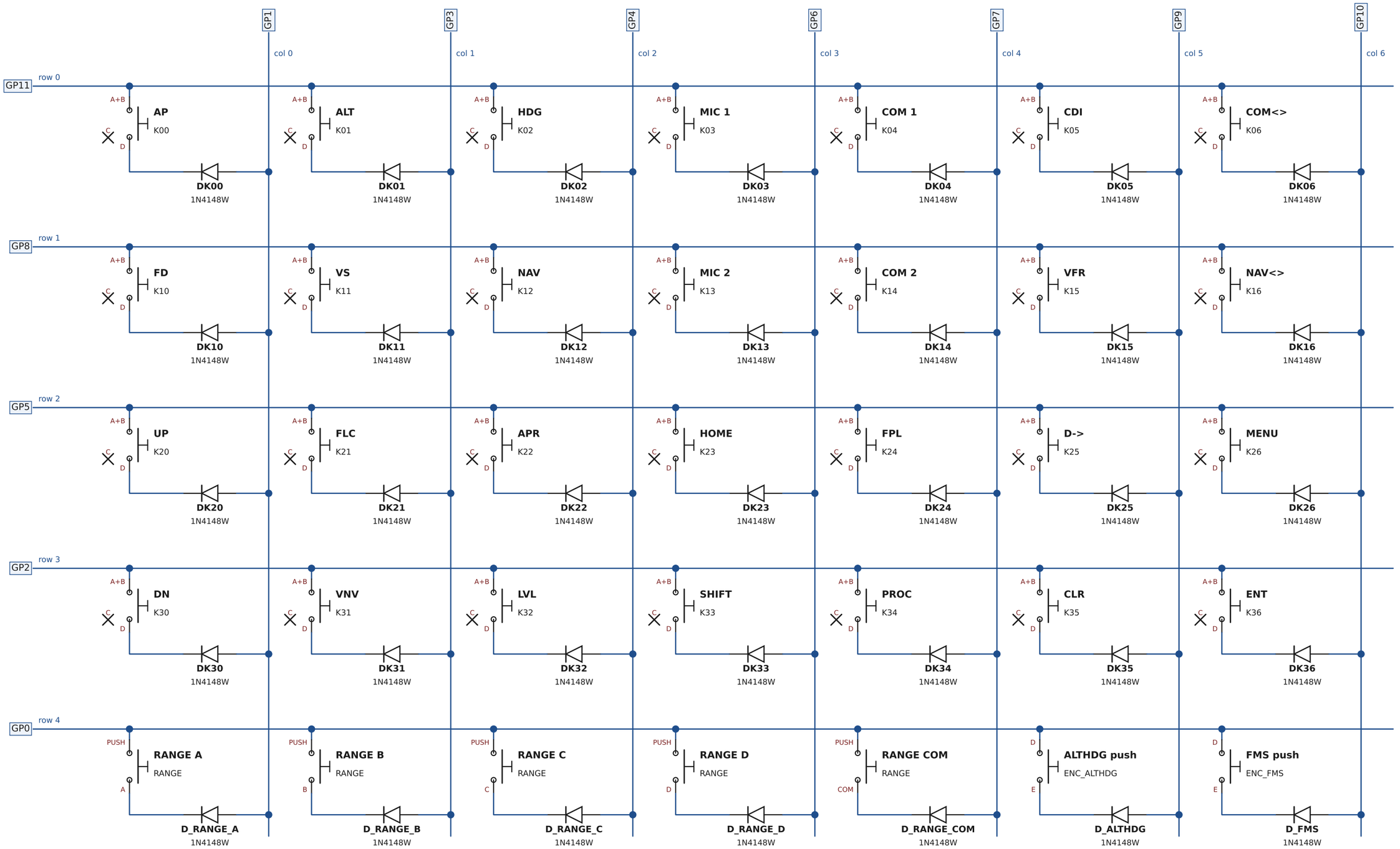
TESTEN (pin 19) goes to GND at the exposed pad.

GPIO14 (GP14) is the spare: a fan-out pad, no load.

GND for an SWD probe: the USB-C shell.

| Avionics Panel rev 1, schematic |
|---|
| Sheet 2 of 4: RP2040, flash, crystal |
| Drawn from the netlist of pcb/board.py (make_schematic.py) |

**Key matrix: rows driven, columns read through the diodes (anodes on the columns: keypad.KeyMatrix(columns_to_anodes=True))**




28 keys 1-1825910-4 (C589907): A and B are one contact, C and D the other;

C is unused. 35 diodes 1N4148W (C81598), one per switch.

Row 4: the RANGE tilt switches A to D and com (centre push = com alone, decoded in firmware),

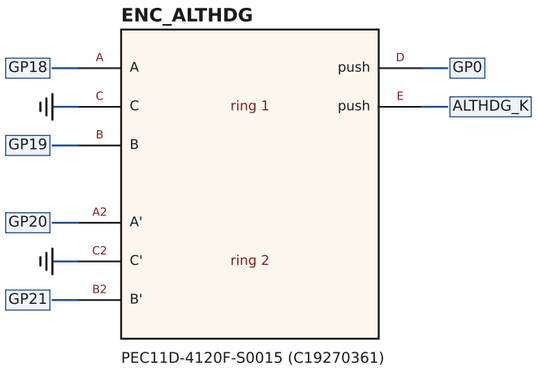
and the ALT/HDG and FMS knob pushes.

**Avionics Panel rev 1, schematic**

Sheet 3 of 4: Key matrix

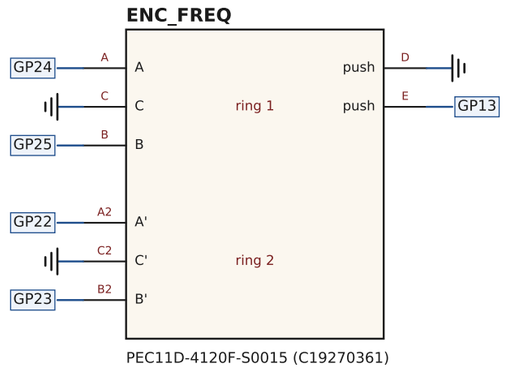
Drawn from the netlist of pcb/board.py (make_schematic.py)

**Dual encoders, 30 detents / 15 pulses with push; the firmware reads ring 1 as the outer and ring 2 as the inner knob**



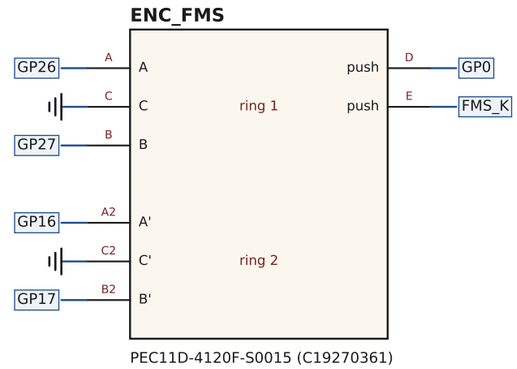

PEC11D-4120F-S0015 (C19270361)

push: key matrix through D_ALTHDG (sheet 3)




PEC11D-4120F-S0015 (C19270361)

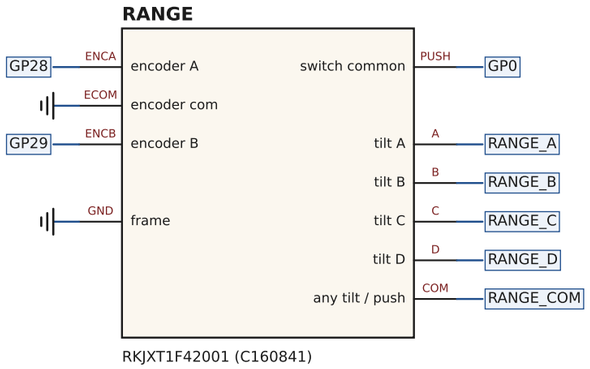
push: direct, GP13 to GND




PEC11D-4120F-S0015 (C19270361)

push: key matrix through D_FMS (sheet 3)

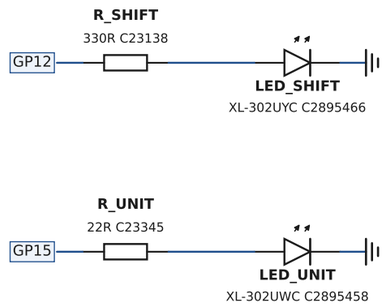
**RANGE: turn, 4-way tilt, push**




RKJXT1F42001 (C160841)

The tilt and push switches sit in key matrix row 4 (sheet 3).

**Lights (light pipes in the front plate)**




**Avionics Panel rev 1, schematic**

Sheet 4 of 4: Knobs and lights

Drawn from the netlist of pcb/board.py (make_schematic.py)